

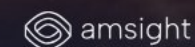
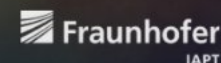
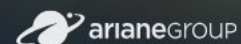
# Materials built for the extreme.

Designed with AI. Made and tested in Europe.

FUNDED BY THE EUROPEAN SPACE AGENCY

Initial deployment of PRISM, for its first use cases: ESA  
FLPP, FIRST! Simulation & Intelligence.

PRISM ALPHA, WITH



Vulcan 2 on test stand P5, DLR Lampoldshausen. Photo: DLR, CC BY 3.0.

## THE PROBLEM

# Far too many possible alloys to make them all.

## 474,311,376

possible alloys from five of nine high-melting metals, mixed in 1% steps

## 130,000 years

to make each one once, at ten a day

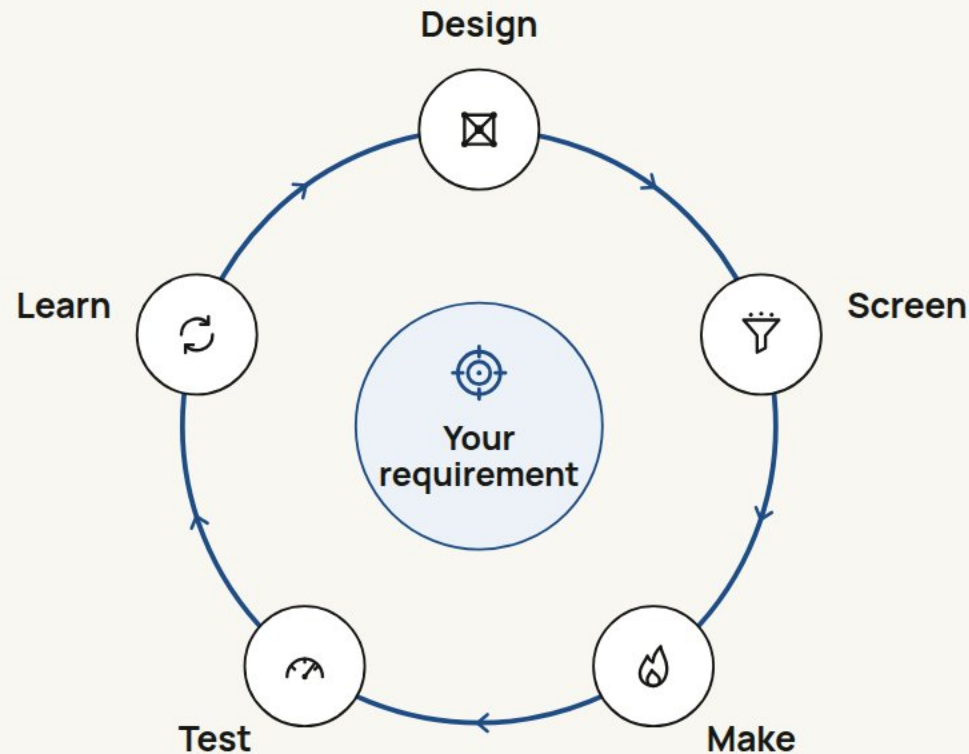
## 10–20 years

for a new material to reach service today

**Sources:** Arithmetic:  $126 \text{ ways to pick five of nine metals that melt above } 1,650^\circ\text{C} \times 3,764,376 \text{ mixes in whole percent}$ . The field is to scale; its layout is illustrative.



# One loop: design, make, test, learn.

**Requirement**

You tell us what the part must survive.

PROTOTYPE

**Design**

AI suggests new mixes, from the whole range.

PROTOTYPE

**Screen**

Simulations rule most out, before anything is melted.

PROTOTYPE

**Make**

We melt and 3D-print the best ideas.

IN USE

**Test**

Each sample is measured against your targets.

IN USE

**Learn**

Every result goes back into the models.

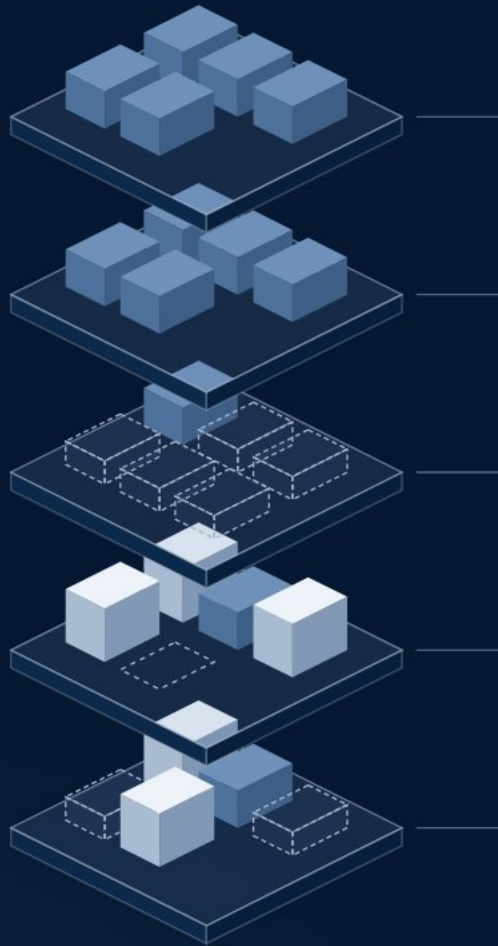
PROTOTYPE

**FIRST TARGET**

Refractory high-entropy alloys for oxygen-rich preburners in rocket engines, to replace Monel K500 and other legacy alloys.

# Five stacks, one system.

■ In use ■ Prototype □ In development --- Goal



## 01 Research

AI suggests mixes; physics simulations throw out what cannot work.

PROTOTYPE

## 02 Harness

Plans each round, runs the tools, scores the results, remembers failures.

PROTOTYPE

## 03 Autonomy

Robots weigh, melt and measure. People stay in charge.

IN DEVELOPMENT

## 04 Manufacturing and test

Melting, metal 3D printing and testing, with Bimo Tech and Fraunhofer IAPT.

IN USE

## 05 Evidence

Every result keeps its source, its owner and its rules.

IN USE

**Maturity:** Each block on a plate is one component, as tall as it is built. In use: runs in current programmes. Prototype: PRISM software, being taken from TRL 3 to 4 in PRISM Alpha. In development: being built.

# Four modules, one loop.



**Loop:** The planner sends a campaign; the samplers propose candidates; simulation and the lab test them; results are kept, with sources; what worked and what failed plans the next round. An engineer signs off what leaves.

# Physical synthesis supplies the reward signal the models train on.

## 01 Recipe translation

PRISM outputs target recipes: precursors, temperatures and thermal profiles.

## 02 Robotic synthesis

Robot arms dose powder and move samples through heating profiles.

## 03 Automated characterisation

Diffraction patterns are captured from every synthesised sample.

## 04 Phase identification

Neural networks compare patterns against structure databases to confirm phases.

## 05 Feedback

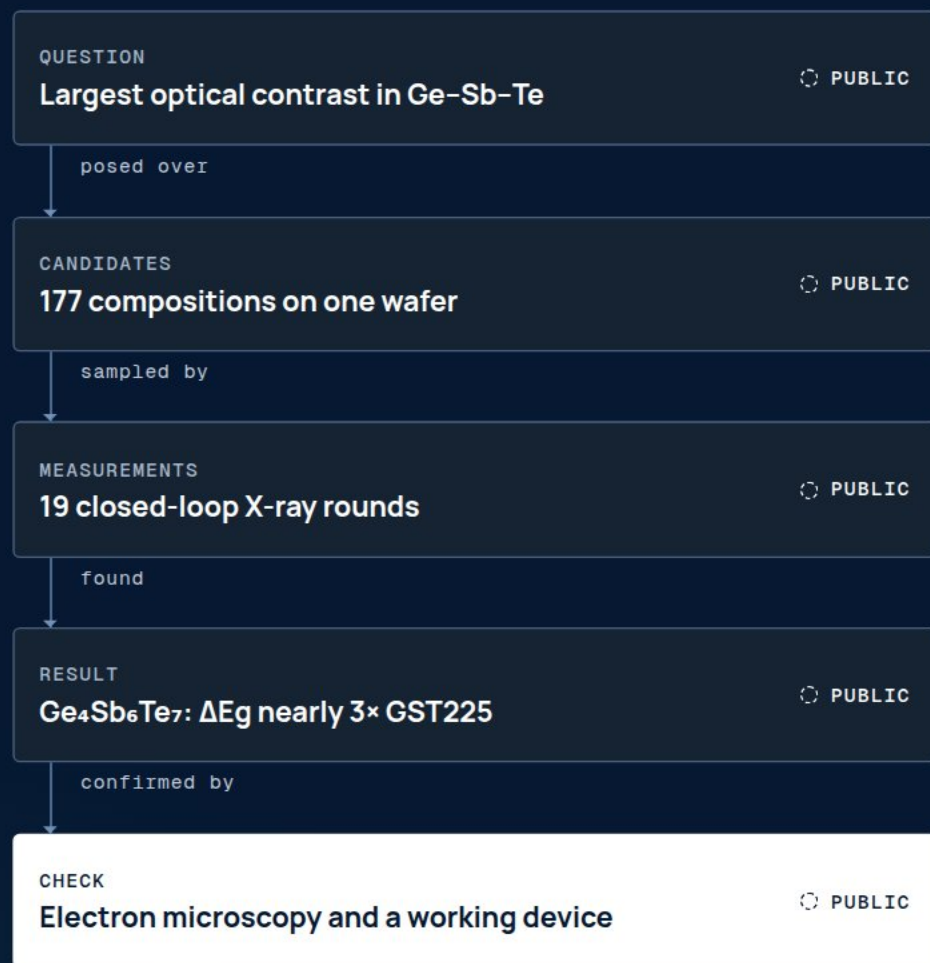
Structured results return to the Evolver's reflector and change the next campaign.

**Maturity:** In development. Direct instrument control closes the loop in phase III of the plan.



The melt spinner in the materials lab at WUST, Wrocław, where our alloys are melted. We rent the equipment.

# Every result carries its own proof.



Partners share data only when they keep control of it. Provenance and rights travel with every result, and derived results inherit the strictest rights of their inputs.

## ■ PRIVATE

Stays with the organisation that produced it.

## ☑ COMPUTABLE

Approved workloads compute on it; nobody inspects it.

## ○ RELEASED

An approved derivative, shared with named parties.

## ⦿ PUBLIC

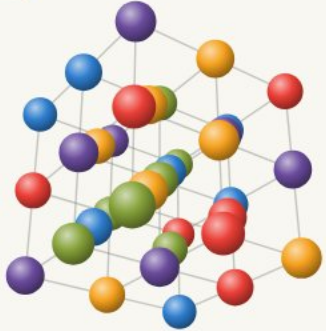
Deliberately published.

**IP: generative sampling conditioned on thermodynamic feasibility reaches metastable alloys that equilibrium screening excludes. Patent filings in progress.**

**Maturity:** Provenance and export classification maintained on ESA work today; machine-enforced rights in development. Lineage shown: NIST's CAMEO, Kusne et al., Nature Communications 11, 5966 (2020), published work by NIST and partners, not PRISM's.

# Funded, contracted and in the lab.

ACTIVE



## Project SPARK

ESA activity · Bimo Tech

# 8

Refractory high-entropy alloy candidates, taken to two physical down-selections.

RUNNING



## PRISM Alpha

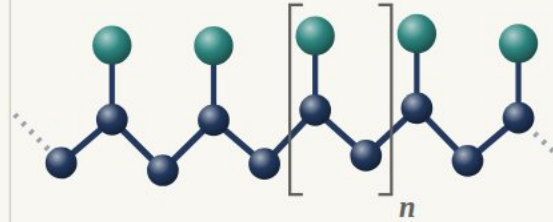
ESA FLPP · FIRST! Simulation & Intelligence

# TRL 3 → 4

In 12 months: at least three candidates and one complete closed loop.

With  arianespace  Fraunhofer IPT  amsight

CONTRACTED



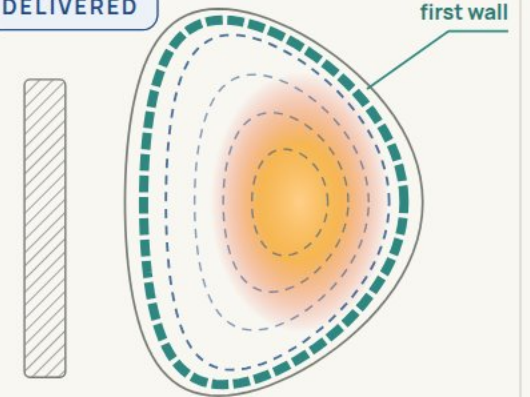
## PFAS-free polymers

Industrial partner · under NDA

# 1st

Privately funded programme, and our first polymer class. Contract and NDA signed; start pending.

DELIVERED



## Fusion heritage

Bimo Tech · ITER

# ITER

Titanium first-wall materials and rhodium targets, supplied by Bimo Tech.

Sources: ESA contract records; Mirdyne and Bimo Tech programme records.

# Where PRISM goes first.

01 · NOW



**Space propulsion**

Alloys for the hottest parts of rocket engines. First with ESA and ArianeGroup.

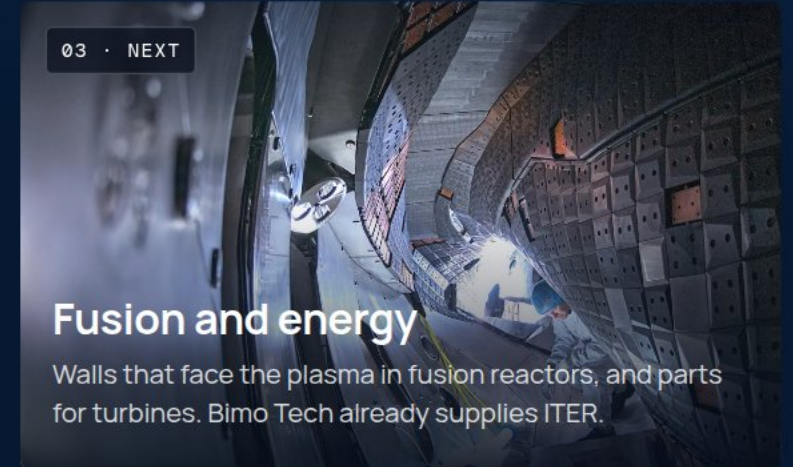
02 · NEXT



**Defence**

Hot-section materials for aircraft engines, with export rules built in.

03 · NEXT



**Fusion and energy**

Walls that face the plasma in fusion reactors, and parts for turbines. Bimo Tech already supplies ITER.

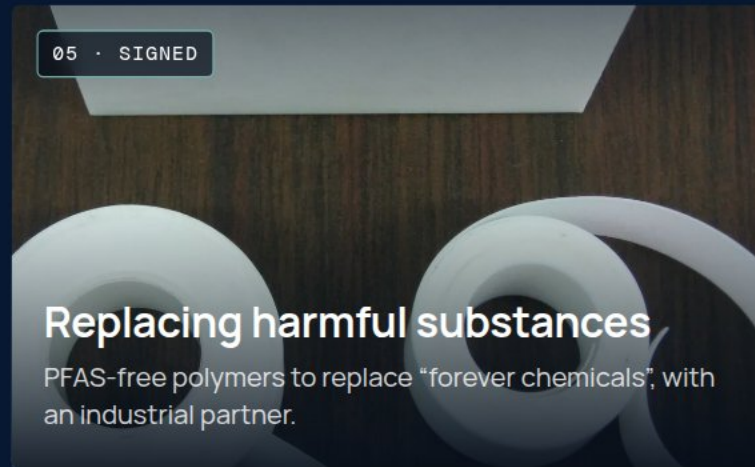
04 · NEXT



**Supply independence**

Alloys that do the job with metals Europe can get.

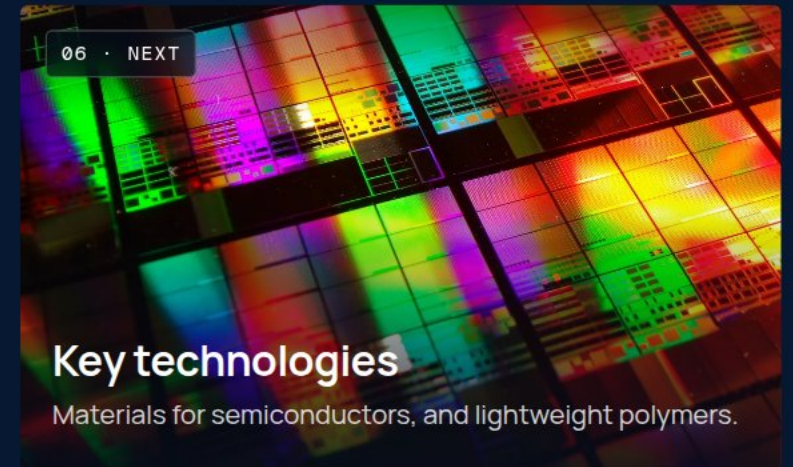
05 · SIGNED



**Replacing harmful substances**

PFAS-free polymers to replace "forever chemicals", with an industrial partner.

06 · NEXT



**Key technologies**

Materials for semiconductors, and lightweight polymers.

**Photos:** Aestus engine in test stand P4.2: DLR, CC BY 3.0 · EJ200 afterburner: Julian Herzog, CC BY 4.0 · Wendelstein 7-X wall tiles: Christopher Roux, EUROfusion, CC BY 4.0 · Tungsten crystals: Alchemist-hp (pse-mendelejew.de), Free Art License. All cropped.

€48M

Serviceable annual wallet

Base case: the upper bound on annual spend across reachable accounts.

€170.7B

EU aerospace  
manufacturing  
turnover  
MKT3

€8.84B

European  
space  
manufacturing  
final sales  
MKT4

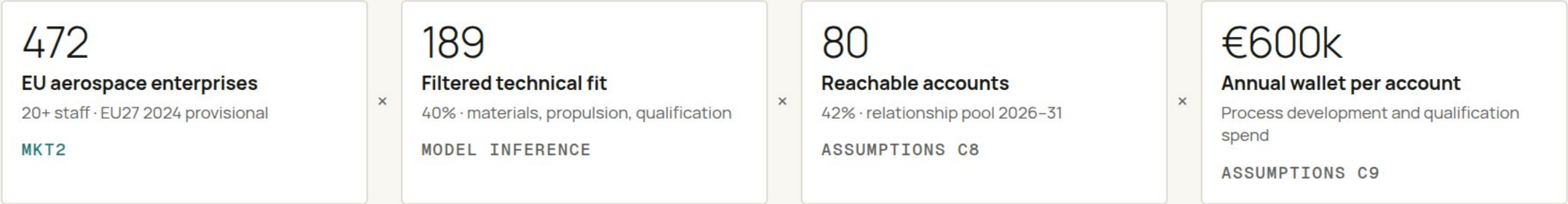
€8.26B

ESA annual  
budget, 2026  
MKT6

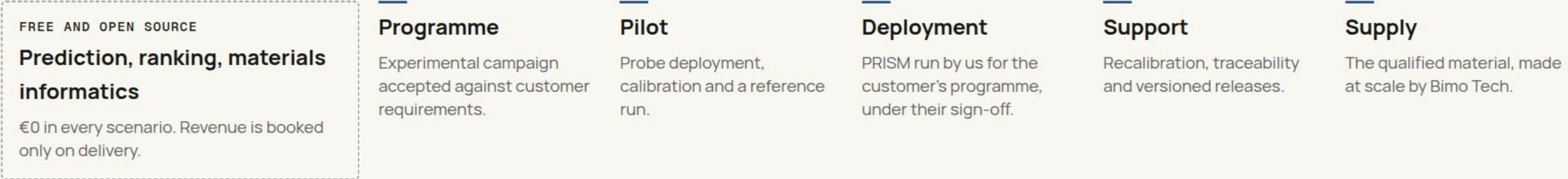
69.6%

Public-  
customer  
share of space  
sales  
MKT5

SERVICEABLE ACCOUNT BRIDGE

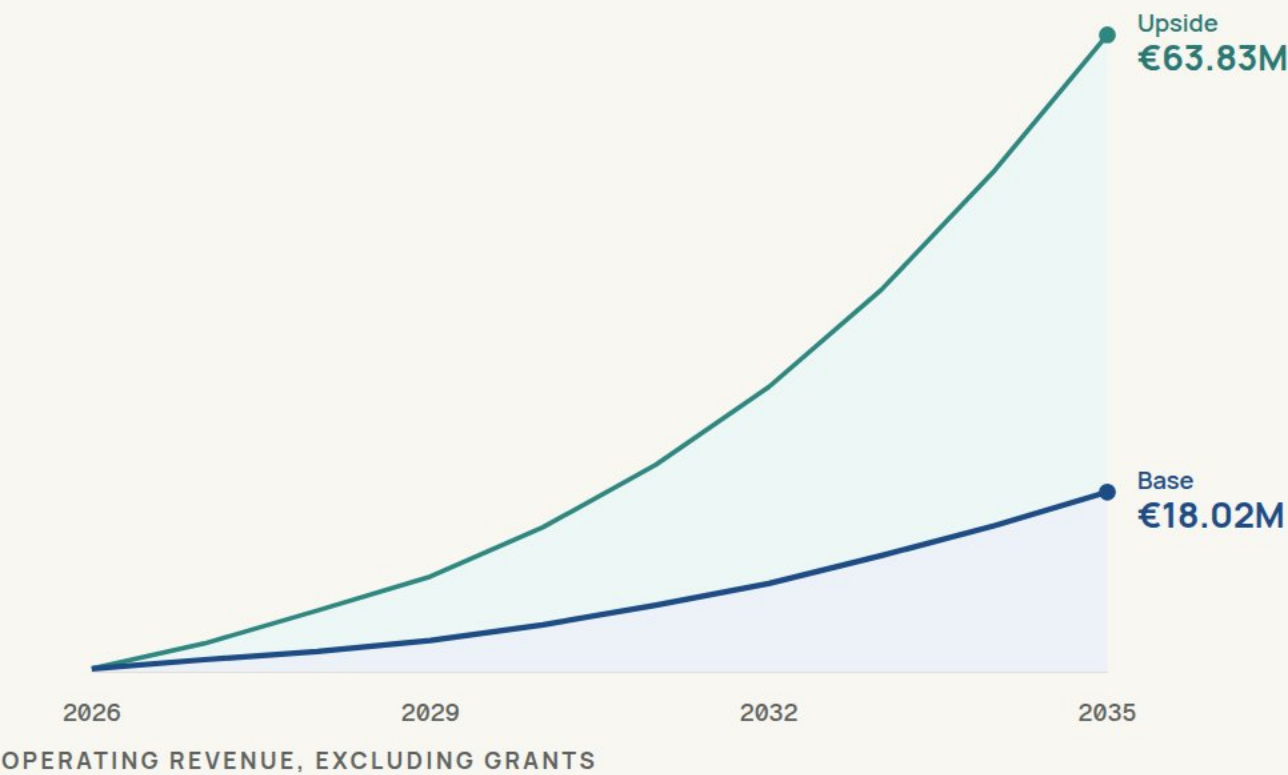


MONETISATION BOUNDARY



Base case: €18.0M revenue by 2035 · EBITDA-positive from 2028 · 61% gross margin. External figures carry their source IDs from the model's evidence register.

# Ten-year operating plan.



|                            | BASE    | UPSIDE  |
|----------------------------|---------|---------|
| Operating revenue 2035     | €18.02M | €63.83M |
| EBITDA 2035, before grants | €4.71M  | €40.84M |
| Gross margin 2035          | 61%     | 81%     |
| EBITDA-positive from       | 2028    | 2027    |
| Cumulative equity required | €2.91M  | €0.58M  |

Revenue is capacity-constrained in both cases: programmes are recognised only up to the delivery capacity left after pilots, deployments and support, so the hiring plan sets the growth rate.

Source: PRISM financial model, research refresh 27 Jul 2026 · checks pass · € nominal.

# The team that runs PRISM's ESA work.

KG

**Kevin Grüning**

MANAGING DIRECTOR, MIRDYNE

Space Systems Lead at Bimo Tech. Physics and technology for space applications, JLU Giessen and THM.

SK

**Siddhartha Yash Kovid**

TECHNICAL LEAD, MIRDYNE

Technical lead of the ESA projects SPARK and PRISM Alpha at Bimo Tech. Applied AI and data science, MIT Professional Education; biomedical engineering, THM.

MO

**Marcin Orzechowski**

CO-FOUNDER, MIRDYNE

CEO and Head of R&D at Bimo Tech, which supplies special metals and precision parts for space, energy and science. Wrocław University of Technology.



**KI-Sonderpreis, Hessen Ideen 2026**

The special prize for artificial intelligence. Photo: Hessen Ideen.



Mirdyne is a spin-off of Bimo Tech, which makes special metals and precision parts and supplies ITER.

# €4M

## Seed round, closing 15 November 2026.

The plan needs €2.91M never to run out of cash. We are raising €4M so the plan survives being wrong, and so the downside gate closes before the money does.

### ALLOCATION

#### Team 65%

Delivery, materials science, product and security headcount

#### Go-to-market, IP and export control 16%

Bids, consortia, field selling, patents, dual-use compliance

#### Compute, infrastructure and lab access 13%

Controlled core, per-site secure compute, partner-lab access

#### Probe hardware, tools and testing 6%

Probe builds, calibration, R&D tooling

### ROADMAP · 24 MONTHS

- I **Computational validation**  
6 months. The full discovery loop running the first PRISM campaigns.
- II **Hybrid loop**  
12 months. Integration with Fraunhofer IAPT additive-manufacturing data.
- III **Autonomous laboratory**  
24 months. Direct instrument control closes the loop with no human in the sequence.

MIRDYNE · GIESSEN, GERMANY

# Start with the capability you need.

Alloys and polymers, designed, made and tested in one loop, with the proof attached.

Register interest →

[mirdyne.com](https://mirdyne.com) →

Initial deployment of PRISM, for its first use cases, funded under ESA FLPP, FIRST! Simulation & Intelligence.

Mirdyne is a spin-off of Bimo Tech. [Technology concept by marc27](#)

